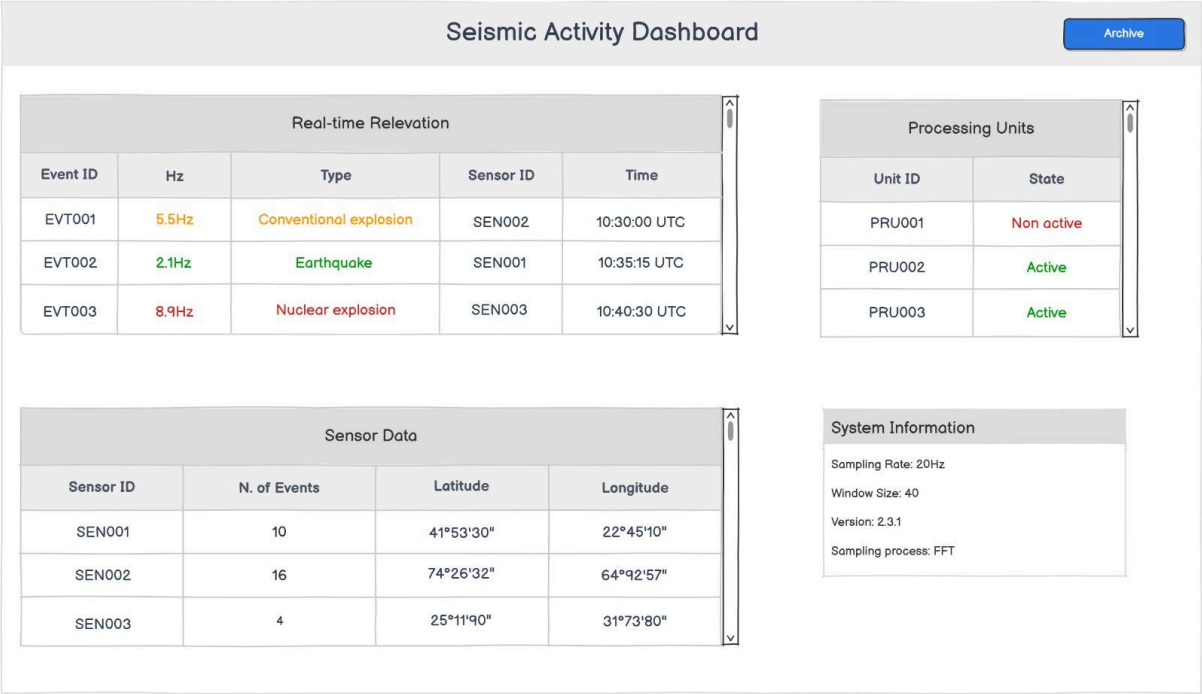
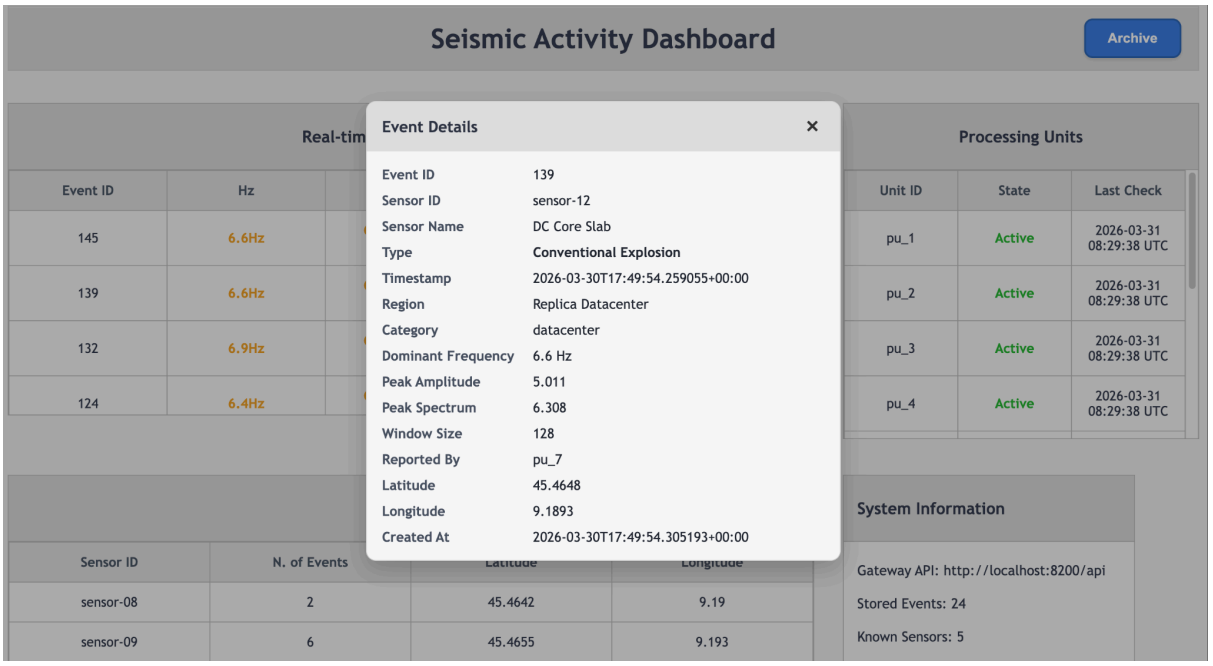
Mockup 1



- **User story 1:** As a user, I want to see the status of all processing units, In order to monitor the operational condition of the system.
- **User story 2:** As a user, I want to see when processing units were last checked, In order to assess the freshness of status information.
- **User story 3:** As a user, I want the processing units status to be refreshed, In order to have a real-time view of them.
- **User story 4:** As a user, I want to monitor seismic activity in real time, In order to immediately notice new events.
- **User story 5:** As a user, I want to see system configuration info, In order to have an overview about the monitoring process.
- **User story 6:** As a user, I want to see only relevant detected events, In order to avoid noise in the dashboard.
- **User story 8:** As a user, I want to have an event classified as an earthquake when the dominant frequency is between 0.5 and 3.0 Hz, In order to easily identify the earthquake event.
- **User story 9:** As a user, I want to have an event classified as a conventional explosion when the dominant frequency is between 3.0 and 8.0 Hz, In order to easily identify the conventional explosion event.
- **User story 10:** As a user, I want to have an event classified a nuclear-like when the dominant frequency is at least 8.0 Hz, In order to easily identify the nuclear-like event.
- **User story 11:** As a user, I want events to be color coded on the dashboard, In order to immediately distinguish the most important detections.
- **User story 12:** As a user, I want to know which sensor or monitored area detected an event, In order to localize the source of the alert.

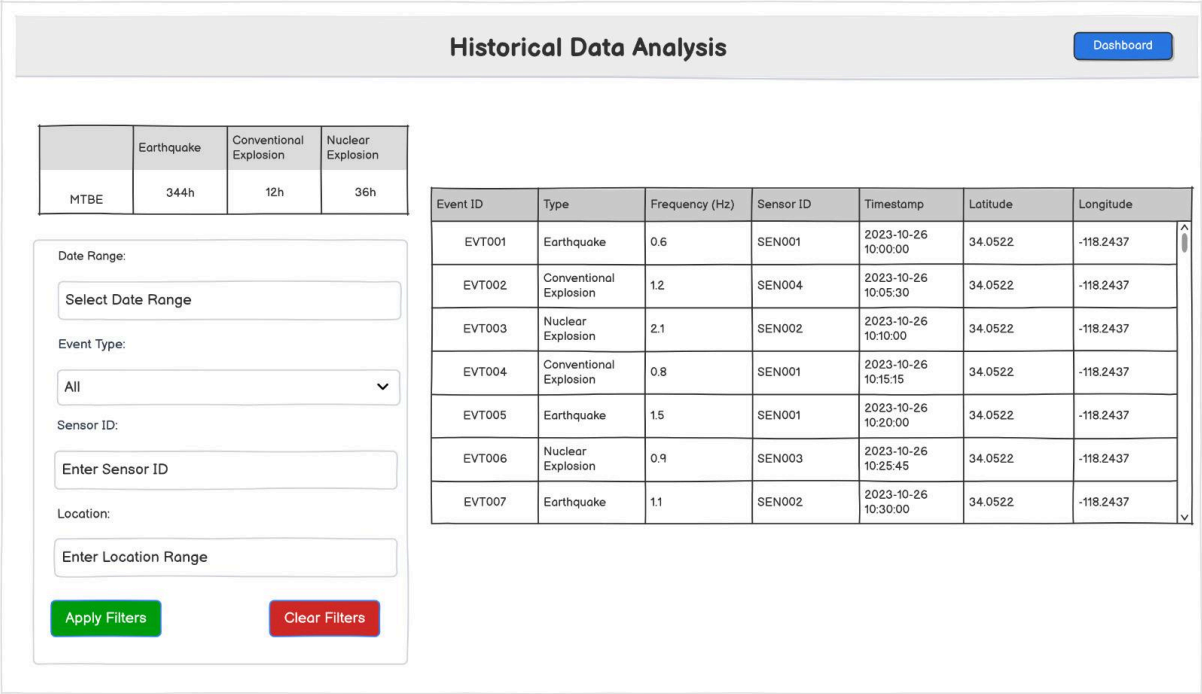
- **User story 13:** As a user, I want to see the list of available sensors, In order to have an overview of the monitored sources.
- **User story 14:** As a user, I want to know the number of detected events for each sensor, In order to quantitatively estimate seismic activity.
- **User story 15:** As a user, I want to know the location of each sensor, In order to identify the monitored area associated with each sensor.

Mockup 2



- **User story 7:** As a user, I want to have a detailed view of a detected event, In order to inspect all the information related to the event.

Mockup 3



- **User story 16:** As a user, I want to see the mean time between earthquake events, In order to estimate how often they happen.
- **User story 17:** As a user, I want to see the mean time between conventional explosion events, In order to estimate how often they happen.
- **User story 18:** As a user, I want to see the mean time between nuclear explosion events, In order to estimate how often they happen.
- **User story 19:** As a user, I want to see historical data about detected events, In order to perform analysis on them.
- **User story 20:** As a user, I want to filter the detected events, In order to have a time-based view.
- **User story 21:** As a user, I want to filter the detected events, In order to have a type-based view.
- **User story 22:** As a user, I want to filter the detected events, In order to have a sensor-based view.
- **User story 23:** As a user, I want to filter the detected events, In order to have a location-based view.

No functional requirements

- **User story 24:** As a user, I want the platform to remain operational even if some processing nodes fail, so that critical monitoring is not interrupted.

- **User story 25:** As a user, I want the system to automatically ingest data from the sensors, In order to have an uninterrupted data stream.